

MA341

APPLIED ODE'S

Final Review

1. Solve the following homogeneous problem $x'(t) = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} x(t)$
2. Solve the following systems of ODEs subject to given initial conditions.

a. $x'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x(t) + \begin{pmatrix} e^t \\ t \end{pmatrix}, x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

b. $x'(t) = \begin{pmatrix} 5 & 3 \\ -6 & -4 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ e^t \end{pmatrix}, x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

c. $x'(t) = \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix} x(t) + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

3. Solve the following problem

$$\begin{cases} y \cos(x+y) dx + (3 \sin(x+y) + y \cos(x+y)) dy = 0 \\ y(0) = \pi/2 \end{cases}$$

4. Consider

$$xy'' - (1+x)y' + y = 0$$

- a. Given $y_1 = e^x$, find the second linearly independent solution.
- b. Use variation of parameters to solve $xy'' - (1+x)y' + y = x^2 e^{2x}$
5. Solve the following initial value problem

$$\begin{cases} y'' - 5y' + 6y = 2\delta(t-1) + 3\delta(t-4) \\ y(0) = 0 \\ y'(0) = 0 \end{cases}$$

6. Solve the following initial value problem

$$\begin{cases} dy + \left(\frac{2y}{x} - \frac{y^3}{x^2}\right) dx = 0 \\ y(1) = 1 \end{cases}$$

7. Consider

$$y'' - 2y' + y = (x+4)e^x$$

- a. Find a general solution.
- b. Find a solution satisfying the ICs $y(0) = 1, y'(0) = 1$.